



VoiceForge Master Studio

Official User Manual & Operational Runbook — v1.3.4

Real-Time Hybrid AI Text-to-Speech Livestream Broadcasting Suite

1. Overview & Architecture

VoiceForge Master Studio bridges live stream chat feeds from multiple broadcasting platforms (Rumble, Kick, X.com, Twitch, YouTube) into synchronized, low-latency neural speech.

Architecture Stack:

- **Native C++ Supervisor:** High-priority launcher utilizing Windows Job Objects to prevent orphaned background tasks.
- **Dual-Engine Audio Dispatcher:** Combines Kokoro-82M (multilingual base) with zero-shot cloning engines (Chatterbox-Turbo, CosyVoice 2).
- **Multicast Scraper:** Headless browser automation with automatic video stream suppression to save system resources.
- **OBS Chroma-Key HUD:** Transparent overlay rendering live chat badges and animated queues.

2. Free vs. PRO Feature Matrix

Feature	Free Version	PRO Version (Gumroad Unlocked)
Kokoro-82M Base Engine	✓ Full Access (54 Voices)	✓ Full Access (54 Voices)
Multilingual Catalogs	✓ 9 Language Packs	✓ 9 Language Packs
Real-Time Speed Modifiers	✓ !name-.75 / !name-1.20	✓ !name-.75 / !name-1.20
Multicast Chat Ingest	✓ All Platforms	✓ All Platforms
OBS Chroma-Key Chat HUD	✓ Included	✓ Included
Speech Moderation & Spam Shield	✓ Included	✓ Included
Chatterbox-Turbo Zero-Shot Cloning	🔒 Locked (Kokoro fallback)	✓ Fully Unlocked
CosyVoice 2 (0.5B) Flow-Matching	🔒 Locked	✓ Fully Unlocked
Qwen3-TTS Multi-Emotion	🔒 Locked	✓ Fully Unlocked
Custom .vfs Voice Audio Files	🔒 Free default voice	✓ Full Neural Voice Cloning

3. Windows 11 Zero-Install Setup

Step 1: Unblock the Downloaded Archive

Right-click `VoiceForgeStudio-Windows-x64.zip` → **Properties** → Check **"Unblock"** → Click **Apply**. Extract the archive.

Step 2: Automated First-Time Setup

Double-click `VoiceForgeStudio.exe` (or `run_voiceforge.bat`). On first run, it automatically launches the native setup engine, downloading embedded Python, PyTorch (CUDA 12.1), and audio libraries. Once completed, the studio window opens automatically.

4. Model Weights & Footprint Management

In the **Settings** →  **Model Weights Manager**, models operate on a dual-engine architecture:

- **Kokoro-82M:** Marked  `Always Active` for all fast multilingual voices.
- **Custom Model Selection:** Select between Chatterbox-Turbo, CosyVoice 2, or Qwen3-TTS. Click  `Activate This Engine` to switch—the previous model deactivates automatically.

- **On-Demand Downloads:** Click  **Auto-Download** to download weights in the background. Progress tracks directly in the top live log bar.
- **Offline / Manual Installation:** Click  **Offline / Manual** to view the exact folder path and direct link to paste files manually on restricted networks.
- **Reclaiming Space:** Click  **Remove** to force-delete the model from both the project folder and Hugging Face cache.

5. Voice Vault & Proprietary .vfs Audio Tracks

VoiceForge recognizes custom voice reference files inside the **voices/** folder:

- **.vfs (VoiceForge Studio Track):** Packaged audio tracks (clean format).
- **.mp3 / .wav**: Standard audio files added by users.

Clicking ► **Preview** on any voice will synthesize audio where the voice announces its active model (e.g., *"Hello! This is Bane, running live on Chatterbox Turbo."*).

6. Chat Commands & Real-Time Speed Modifiers

- **Calling Voices:** Prefix the voice name with an exclamation mark (e.g., **!bane Greetings stream**).
- **Speed Tags:** Append speed modifiers directly to the voice tag:
 - **!bane-.75 Slow speech tempo** (0.75x speed)
 - **!bella-1.20 Fast announcement** (1.20x speed)
- **Multi-Voice Sentences:** Combine multiple voices in a single message:

```
!bane-.75 I am speaking slow. !bella-1.20 And I am speaking fast!
```

7. Moderation, Anti-Spam & Whitelisting

- **Spoken-Text-Only Censorship:** Blocked keywords only apply to spoken message content. Usernames are never blocked or censored.
- **Spam Shield (8-Second Duplicate Window):** If a viewer sends the exact same message twice within 8 seconds, it is sent to the **Filtered Messages Log** table with a  **Restore** button.
- **Whitelisted Users:** Add usernames to the Whitelist in the Audio Log tab to bypass all filters.

8. Troubleshooting Quick-Guide

- **Port 8080 Conflict (Errno 10048):** Run `taskkill /F /IM python.exe /IM VoiceForgeStudio.exe` in Command Prompt to clear stale background processes. `run_voiceforge.bat` performs this automatically on launch.
- **Smart App Control (SAC) Block:** Run `Get-ChildItem -Recurse | Unblock-File` in PowerShell, or launch using `run_voiceforge.bat`.
- **No Sound:** Check if the top button reads  `Click to Enable Audio`. Click it once to activate browser autoplay.